at the reference frequency, and one could explicitly set

$$\tilde{\phi}_0(f)\delta\tilde{\phi}_{\rm rel}(f) \mapsto \left[\tilde{\phi}_0(f) - \tilde{\phi}_0(f_{\rm ref})\right]\delta\tilde{\phi}_{\rm rel}(f) + \tilde{\phi}_{\rm ref}. \quad (14)$$

In this formulation, phase errors would accumulate as one moves away (both in the positive and negative $f$-direction) from $f_{\rm ref}$.

From now on, we will drop the subscript 'rel' from the $\delta\tilde{\phi}$ parameter and we will refer these parameters as the pair $(\delta\tilde{A}, \delta\tilde{\phi})$, which in general are functions of frequency. These parameters will represent either the abs-phase or rel-phase model, depending on the context of which parametrization is being discussed. We refer to these models as waveform error parametrization models or WF-Error parametrization for short.

Following the calibration uncertainty framework, we use N nodal points (or knots) between the frequency interval $f \in [f_{\rm min}, f_{\rm max}]$. Here, $f_{\rm min}$ serves as the low-frequency cutoff, or the frequency for the first nodal point, while $f_{\rm max}$ represents the maximum frequency of our analysis which can be a frequency immediately following the merger. In this way, we can account for waveform errors throughout the signal in the detector band. The values of the WF-Error parametrization parameters at the waveform nodal points can be utilized for cubic spline interpolation. This method employs piecewise polynomial curves that are cubic in order and have continuous second derivatives. We utilize the CubicSpline implementation from the publicly available SciPy package [103] for our WF-Error parametrizations. Using $f_i \in [f_1, f_2, ....f_N]$, which are the nodes of the polynomial in the frequency, we construct $\delta\tilde{A}(f)$ and $\delta\tilde{\phi}(f)$ curves using:

$$\delta\tilde{A}(f) = P_s^3(f; \{f_i, \delta\tilde{A}_i\}) \quad (15)$$

$$\delta\tilde{\phi}(f) = P_s^3(f; \{f_i, \delta\tilde{\phi}_i\}) \quad (16)$$

where $P_s^3$ is the cubic spline polynomial. In the PE analysis, we provide the priors for $\delta\tilde{A}_i$ and $\delta\tilde{\phi}_i$ at each frequency node. Each realization of $(\delta\tilde{A}_i, \delta\tilde{\phi}_i)$ points corresponds to the curves $\delta\tilde{A}(f)$, and $\delta\tilde{\phi}(f)$ which is used to modify the reference waveform model $\tilde{h}_{ref}(f)$ in accordance to equations (12) and (13).

We now turn to the question why we introduce two different phase parametrizations. The SNR (3) and the inner product (A2) used in PE obey a symmetry between the observed data and the GW model. Therefore, if one accounts for calibration uncertainties in the data, this should have the same effect as accounting for waveform systematics. Since the calibration uncertainties are modeled with absolute errors in phase, marginalizing over calibration uncertainties can account for waveform systematics if they are of the same order. However, it is worth pointing out that all GW Transient Catalogs (GWTCs) included the effect of uncertain detector calibration in the analysis of GW events. Despite marginalizing over amplitude and (absolute) phase uncertainty for each detector, the properties of some binaries showed significant systematic differences between the waveform models that have been employed for the analysis (e.g., GW190412 [104], GW191109_010717, GW191219_163120, GW200129_065458; see Sec. III E in [6]). Evidently, marginalizing over calibration uncertainties did not incorporate all waveform systematics.

This is not unexpected. Systematic uncertainties in waveform models can often appear as secular phase drifts caused by numerical errors, (unavoidably) incomplete series expansions, interpolation inaccuracies, or even missing physics. The resulting waveform errors can, therefore, become much more severe than random small phase variations. If one wants to allow for possibly significant waveform uncertainties, large absolute phase errors must be considered, which may lead to an overly pessimistic error estimate. Alternatively, a small relative phase error is likely more appropriate to model systematic waveform uncertainties at the leading order.

We now turn to incorporating WF-Error parametrizations in PE analysis. Ideally, a waveform model shall consist of the best-estimated reference model $\tilde{h}_{\rm ref}$, and priors for the parameters $\delta\tilde{A}$ and $\delta\tilde{\phi}$ for rel-phase or abs-phase parametrization to account for the waveform systematics. We call them WF-Error priors or error budget. The WF-Error priors should generally depend on the parameter space. We expect these priors to be narrower in the regime where many NR simulations are available, and waveform models are calibrated to greater accuracy. For the regions where NR simulations are sparse and theoretical waveform description is not so accurate, we expect them to be broader. For a given signal in parameter space, we also expect these priors to be broader for the frequencies close to the merger, where the analytical models extending from the inspiral regime to the merger are calibrated with NR simulations.

In the absence of such priors from the waveform modeling side, we can use a conservative approach and use wide enough priors so that we expect to capture any systematic and use data to constrain the parameters related to waveform errors. It might lead to broader posterior distribution resulting in larger variance on signal parameters (especially for weak signals), but it is expected to be unbiased. We will test these assumptions in the section V with simulated signals and recovery.

Figure 2 shows an example of waveform with GW150914 type signals and the modifications applied in the frequency domain using cubic splines. To generate the reference signal, we use the `IMRPhenomPv2` waveform model with source frame masses $(m_1^{\rm src}, m_2^{\rm src}) = (36M_\odot, 29M_\odot)$ at the luminosity distance of 500 Mpc. We used non-spinning injections for this example. For the modifications, we use both the parametrizations described previously with WF-Error parameters chosen from a realization of the Normal distribution described in the figure. For the same values of $\delta\tilde{\phi}$, the rel-phase parametrization significantly modifies the reference model compared to abs-phase parametrization. In Figure 3, we show the reference waveform model for the GW150914 type signal described above along with the modified waveform curves with abs-phase

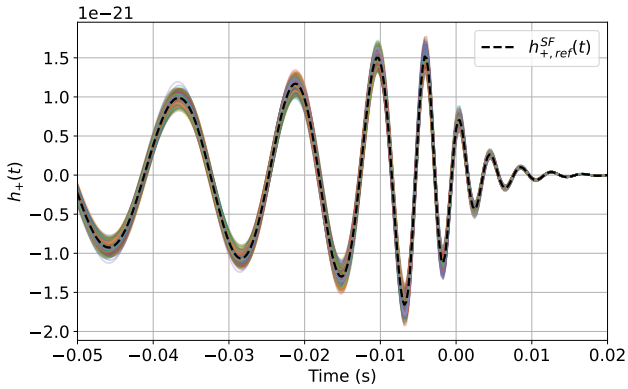


FIG. 3. We show the time domain waveform for (+) polarization for the GW150914 type signal. The reference waveform model `IMRPhenomPv2`: $\tilde{h}^{SF}_{+,ref}(t)$ is shown as dashed black curve. Other curves are modified waveform with abs-phase modification, described by equation (12), with WF-Error parameters sampled from the distribution $\delta\tilde{A}, \delta\tilde{\phi} \sim \mathcal{N}(\mu = 0, \sigma = 0.05)$.

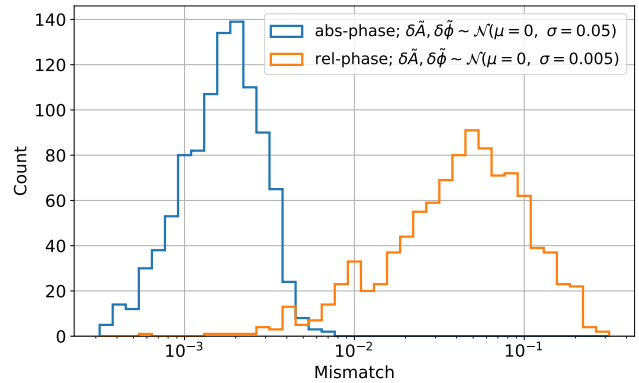


FIG. 4. On the x-axis, we show the mismatch between the reference waveform model $h_{\rm ref}$ and the modified waveform model $h_{\rm mod}$. The modified signal, $h_{\rm mod}$, is generated by either abs-phase or rel-phase WF-Error parametrization. The WF-Error parameters are taken from the normal distribution, and we generate cubic spline curves to modify the reference model. The rel-phase model is generally well suited to model comparatively more significant mismatches.

modification. We use cubic splines with ten waveform nodal points in frequencies with values $(\delta\tilde{A}, \delta\tilde{\phi})$ drawn from a normal distribution $\mathcal{N}(\mu = 0, \sigma = 0.05)$.

It is important to note that the WF-Error parametrizations (abs-phase and rel-phase) can be transformed into one another using simple mathematical relations by evaluating the phase $\tilde{\phi}_0$ of the reference waveform model $\tilde{h}_{\rm ref}$. We can use one of the parametrizations and mathematical transformations to convert it into the other according to the situation. However, using the correct parametrization is desirable for practical purposes in PE analysis. As an example, if the rel-phase parametrization is used for waveform systematics, and if one chooses abs-phase parametrization in PE, the priors for $\delta\tilde{\phi}$ need to become wider for larger frequencies to account for the growing reference phase $\tilde{\phi}_0$.

We can use hints from the mismatch (see Appendix A for definition) studies to decide which WF-Error parametrization is better suited for practical purposes. In Figure 4, we show the distribution of mismatches between the reference waveform model and the modified one using both parametrizations. We use the Gaussian prior with zero mean and a fixed standard deviation across frequency range to generate the modified signal $h_{\rm mod}$. As expected, even with the narrower Gaussian prior, the rel-phase parametrization produces significant deviation to the $h_{\rm ref}$ compared to the abs-phase model, hence more significant mismatches. When we are in the parameter space where larger mismatches are expected, we can set our PE analysis framework to use rel-phase parametrization.

### C. Degeneracy between detector calibration and waveform uncertainties

A comprehensive PE data analysis pipeline for PE should include i) methods to incorporate uncertainties in waveform modeling, ii) corrections due to detector calibration, and iii) adjustments for possible deviations from Gaussian and stationary noise, whenever applicable. If we choose cubic-spline curves to mitigate waveform uncertainties and detector calibration uncertainties, a natural question arises: can these systematics be decoupled? We will explore the specifics of this query in a future study. However, we would like to point out that there are fundamental differences in how these corrections are applied: While the calibration corrections are applied in the detector frame and are different for each detector, the WF-Error parametrization corrections are applied before projecting the waveform into each detector. Therefore, the modification due to the calibration uncertainties and abs-phase waveform errors can be decoupled for a sufficiently loud signal.

## IV. CAN FISHER MATRIX ESTIMATES BE USED FOR ACCOUNTING SYSTEMATIC BIASES?

A standard tool in the literature to quantify systematic biases between two waveform models is the Fisher matrix formalism [28, 105]. In this section, we will apply it to see how significantly errors modeled as (13) change the recovery of source parameters $\Theta^\mu$ and verify whether the formalism can account for the induced biases. If it can, this will help us predict the systematic parameter errors induced by externally provided estimates of wave-

form errors. Then we can assess the significance of these errors and determine if corrections are necessary. For the science case for next-generation GW detectors, many investigations focus on forecast studies using Fisher matrix formalism [106–108]. This section specifies the conditions under which corrections based on Fisher matrix analysis are applicable or not. Appendix A introduces the Fisher matrix formalism.

### A. Setup of the analysis

In this part, we introduce methods to test how well the Fisher estimates work in trying to recover biases induced by certain choices of $\delta\tilde{A}, \delta\tilde{\phi}$. We do that by injecting data generated from the signal model (13) for chosen values $\delta\tilde{A}, \delta\tilde{\phi}$, and recovering using the baseline waveform model. The injection parameters chosen for these analyses are quoted in table I. We focus on non-spinning binaries and keep the extrinsic parameters $\psi, \alpha, \delta$ fixed, so in the end only $\mathcal{M}^z, q, D_L, \iota$ and time of coalescence $t_0$, global phase $\phi_0$ are varied. Accordingly, the set of parameters constituting the Fisher matrices we calculate is

$$\{\mathcal{M}^z, q, D_L, \iota, t_c, \phi_0\}\,. \tag{17}$$

Please refer to Appendix A for details on the calculation.

There are two properties of the posteriors that we wish to compare to the respective Fisher matrix estimate:

1. First of all, the systematic error $\Delta\Theta^\mu$. From (A3) we see that it is formally defined as the difference between maximum a posteriori estimate $\Theta_{\rm bf}$ and injected parameters $\Theta_{\rm tr}$.

   In practice, we follow the method chosen in the gravitational-wave transient catalogs [6, 109–111], and use median values as proxies for the maximum a posteriori points.

2. Secondly, the posterior width. Ref. [112] shows that, in the high-SNR limit, it reads

$$\sigma_{\Theta^\mu} \simeq \sqrt{(\Gamma^{-1})^{\mu\mu}} \tag{18}$$

   and this number is compared to the standard deviation of the marginalized posterior for $\Theta^\mu$.

### B. Results of comparing Fisher matrix estimates with PE runs

In total, we have analyzed four scenarios: two different sets of error values of $\delta\tilde{A} = 0.004, \delta\tilde{\phi} = 0.004$ and $\delta\tilde{A} = 0.01, \delta\tilde{\phi} = 0.01$, and for each of them one moderate- and one high-SNR case (cf. table I). All of the signals have been injected as a zero-noise injection into the Hanford detector with an O4 PSD. The results are summarized in table II.

Since the chosen signal lies in the non-spinning part of the parameter space, the remaining intrinsic parameters are the masses parametrized by the tuple $\mathcal{M}^z, q$. We will direct our attention toward these two, not discussing biases in the luminosity distance $D_L$ and inclination $\iota$ in detail. This is because the uncertainty in $D_L$ is typically very large. Moreover, there is a known correlation between these two parameters, which makes their recovery and the assessment of their bias results more complicated. We have verified that this correlation has no significant impact on results for other parameters.

The first important finding is that the PE results are biased if a modified signal is recovered using the baseline, non-modified model. This finding implies that the error model (13) is capable of parametrizing systematic errors in the way we intend it to. Moreover, we find that these biases can be recovered in the Fisher matrix framework, validating that the biases have a physical origin rather than being an artifact of flawed application of PE or an error in the implementation. The statistical bias is recovered very well in all four scenarios, with differences usually appearing in the second significant digit. This agreement is an interesting feature of the results because it indicates that the width estimate (A5) seems to be more robust to increasing waveform differences than the systematic bias estimate (A3) (at least for our applications), where only the first configuration $\delta\tilde{A} = \delta\tilde{\phi} = 0.004$ shows consistency between Fisher matrix estimates and PE results.

Those results motivate the following interpretation: in case of $\delta\tilde{A} = \delta\tilde{\phi} = 0.004$, the waveform differences introduced into the waveform are still well approximated by the linear expansion (A6), the LSA is justified; this clearly changes for $\delta\tilde{A} = \delta\tilde{\phi} = 0.01$, where the systematic bias estimates begin to diverge from the corresponding PE results – the LSA is not a faithful approximation anymore. This divergence happens in various ways, with both magnitude and sign being sources of inconsistency.

We have seen that Fisher estimates can work, but not always. First of all, it was expected that the LSA will not hold up to arbitrarily large values of $\delta\tilde{A}, \delta\tilde{\phi}$, simply because we use the relative error model where errors accumulate from low to high frequencies and thus will quickly result in large waveform differences. However, this being the case for a 1% error and the simplest setup we can think of (constant shifts, same deviations in both polarizations) has serious implications for our objectives of a practical application of the Fisher estimates: for realistic scenarios, where errors of this magnitude are expected, they are not a suitable prospect to predict systematic errors in intrinsic parameters induced by $\delta\tilde{A}, \delta\tilde{\phi}$. Of course, in other parts of the parameter space, other thresholds for the LSA validity might exist, but the existence of such a counterexample in a part of the parameter that is certainly relevant for detected signals is already sufficient to recognize that we should not rely on a faithful estimation of biases. The numbers are particularly concerning for the high-SNR signal with $\delta\tilde{A} = \delta\tilde{\phi} = 0.01$ since this